

AI Researcher Collaboration Platform

Developer Brief — MVP

1. Project Overview

We are building a bilingual Arabic/English AI Researcher Collaboration Platform MVP from scratch.

The client has an Excel/CSV dataset of approximately 70 researchers. An external researcher or university user will be able to describe a research need/challenge in natural language, and the system will use the OpenAI GPT API to match the requirement against the existing researcher dataset and recommend suitable individual researchers or research teams.

The user can then select a recommendation and submit a collaboration request, which will be reviewed by the Research Center Director.

The MVP should be clean, professional, responsive, and complete enough for end-to-end client testing.

2. User Roles

There are 2 roles in the MVP:

Standard User

An external researcher/university user who can:

- Register/login to the platform.
- Submit a research need/challenge in Arabic or English.
- Receive individual researcher/team recommendations.
- View recommended researcher information.
- Select a researcher/team.
- Submit a collaboration request.
- View their submitted requests and current statuses.

Research Center Director

The Director can:

- Login with authorized Director access.
- Access the Director dashboard.
- View submitted collaboration requests.

- Review the original research need and selected/recommended researcher/team.
- Approve, Reject, or Request Changes.
- Add a note when requesting changes.
- Track request statuses.

No additional roles are required for this MVP.

3. Researcher Dataset & Profiles

The client will provide approximately 70 researcher records in Excel/CSV containing:

1. Full Name
2. Email Address
3. Academic Position
4. Primary Research Areas
5. Specific Research Interests
6. Keywords / Tags
7. Current Research Projects
8. Past Research Projects
9. Google Scholar Profile URL
10. Number of Publications
11. Types of Opportunities Interested In
12. Preferred Project Duration
13. Availability Status
14. Willingness to Lead Projects
15. Technical Skills
16. Additional Skills
17. CV / Resume
18. Website
19. LinkedIn Profile
20. Brief Bio / About Me

The dataset should be imported and stored in the platform.

Users should be able to view the relevant information of recommended researchers through a clean researcher profile/detail view. Only available fields need to be displayed.

4. Research Need / Challenge Submission

The Standard User can describe the researcher/expertise they need using natural language.

Example:

“I need researchers experienced in AI and healthcare who are available and willing to lead a short-term research project.”

The research need can be submitted in Arabic or English.

5. AI Researcher & Team Matching

The OpenAI GPT API will be used to understand the submitted research need and match it against the client's existing researcher dataset.

Matching should consider relevant available information such as:

- Research areas and interests
- Keywords/tags
- Technical/additional skills
- Current and past projects
- Availability
- Preferred project duration
- Willingness to lead
- Opportunity interests

Individual Recommendations

Recommend suitable individual researchers and provide a brief reason explaining why each researcher is relevant to the submitted requirement.

Team Recommendations

Where the research requirement benefits from complementary expertise, the system should also recommend a suitable team of researchers whose combined expertise fits the requirement.

A team recommendation does not need to be forced where an individual researcher sufficiently matches the requirement.

All recommended researchers must come from the client's provided dataset.

Important: No semantic/vector search, embeddings, or vector database is required. The client specifically simplified the requirement to use the GPT API only because the dataset is small.

6. Collaboration Workflow

After reviewing the recommendations:

Select Researcher/Team → Submit Collaboration Request → Director Review

The Director can:

- Approve
- Reject
- Request Changes

Main statuses:

Pending / Approved / Rejected / Changes Requested

When requesting changes, the Director can add a short note explaining what needs clarification/change. The user can see the note with the updated status.

A separate chat/messaging system is not required.

7. Research Center Director Dashboard

The Director dashboard should allow the Director to:

- View submitted collaboration requests.
- Open request details.
- View the original research need/challenge.
- View the selected/recommended researcher or team.
- Approve or reject requests.
- Request changes with a note.
- View and manage request statuses.

No advanced analytics dashboard is required.

8. Authentication & Authorization

Authentication and authorization are included.

The platform should have:

- Standard User registration/login.
- Research Center Director login/access.
- Role-based permissions.
- Director functionality protected from Standard Users.

9. Arabic & English

The MVP should support:

- English interface.
- Arabic interface.
- Proper Arabic RTL layout.
- Research queries in Arabic and English.

Researcher information can remain in the language provided in the client's dataset. Automatic translation of researcher data/CVs is not required.

10. Main Screens

Approximately 6 core areas/screens are expected:

1. Login / Registration / Authentication
2. Researcher Data & Profile View
3. Submit Research Need / Challenge
4. AI Researcher / Team Recommendations
5. Collaboration Request & Status
6. Research Center Director Dashboard / Request Management

Supporting forms, detail views, or modals required for these features can be added as needed.

11. Complete Flow

Standard User

Register/Login → Submit Research Need → GPT analyzes requirement → Researcher dataset matched → Individual Researcher/Team Recommendations → Review Recommendations → View Researcher Details → Select Researcher/Team → Submit Collaboration Request → Pending → Director Review → Approved / Rejected / Changes Requested → User sees updated status

Director

Login → Director Dashboard → View Request → Review Research Need + Selected/Recommended Researchers → Approve / Reject / Request Changes → Status Updated for User

12. Technical Stack

Frontend: React.js

Backend: Node.js + Express.js

Database: PostgreSQL / Supabase

Authentication: Supabase Auth

AI: OpenAI GPT API

Deployment: Vercel frontend + suitable backend hosting

Version Control: Git/GitHub

The repository must remain private.

The completed MVP should be deployed to a staging/testing URL for client testing and should also be runnable locally.

13. Not Included in This MVP

- Semantic/vector search

- Embeddings/vector database
- Custom AI/ML model training
- Automatic Google Scholar scraping/synchronization
- External university/company system integrations
- Native mobile application
- Advanced analytics/reporting
- Email/SMS/push notifications
- Internal chat/messaging system
- Automatic translation of researcher data/CVs
- Paid production infrastructure
- Additional functionality outside the agreed MVP

14. Important Client Notes

- The client has approximately 70 researchers, provided through Excel/CSV.
- Matching should work as: External researcher/university query → existing researcher dataset → relevant researcher/team recommendations.
- The client specifically removed the original semantic-search requirement and requested GPT API only.
- There is no existing application/codebase; the project is being built from scratch.
- The client expects a complete MVP with UI, core functionality, dashboards, and end-to-end workflow, not only an AI matching prototype.
- The entire project is private and confidential. Do not publicly share the project, dataset, screenshots, organization information, or source code.

Expected Final Result

The final MVP should provide this complete working flow:

User submits a research need in Arabic/English → receives suitable individual researcher/team recommendations from the existing dataset → selects a recommendation → submits a collaboration request → Research Center Director reviews it → Approves / Rejects / Requests Changes → user sees the updated status.

The final application should be clean, professional, responsive, secure, and fully testable end-to-end.